Antonio Alcántara Mata

Tenure-track Assistant Professor

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ACADEMIC APPOINTMENTS

Sep. 2026 – Present | **CUNEF Universidad** | *Spain*
Tenure-track Assistant Professor, Quantitative Methods Department
Researching on machine learning and data-driven optimization for decision-making under uncertainty, with a particular interest in energy/power systems, alongside teaching courses in analytics and operations research.

Sep. 2025 – Aug. 2026 | **Technical University of Denmark (DTU)** | *Denmark*
Postdoctoral Researcher, Department of Wind and Energy Systems
Advisor: Prof. Spyros Chatzivasileiadis
Foundation models for power systems, with a focus on uncertainty quantification and trustworthiness.

Dec. 2021 – Jun. 2025 | **Universidad Carlos III de Madrid (UC3M)** | *Spain*
Predoctoral Researcher (FPU Fellow), Department of Statistics
Advisor: Prof. Carlos Ruiz Mora
Machine learning and operations research, focusing on the use of ML models as surrogates in optimization problems, such as neural network MILP reformulations.

Jan. 2021 – Dec. 2021 | **Universidad Carlos III de Madrid (UC3M)** | *Spain*
Research Assistant, Department of Computer Science
Advisor: Prof. Inés M. Galván
Probabilistic forecasting models for regional renewable energy generation.

RESEARCH STAYS

Aug. 2024 – Oct. 2024 | **KTH Royal Institute of Technology** | *Sweden*
Visiting Researcher, Department of Mathematics
Host: Prof. Jan Kronqvist

Sep. 2023 – Dec. 2023 | **Imperial College London** | *United Kingdom*
Visiting Researcher, Department of Computing
Host: Prof. Calvin Tsay

EDUCATION

May 2025 | **Universidad Carlos III de Madrid (UC3M)** | *Spain*
Ph.D., Mathematical Engineering
Advisor: Prof. Carlos Ruiz Mora
Dissertation: “Machine Learning for Decision-making under Uncertainty: New Methodologies and Energy Applications”

Jul. 2021 | **Universidad de Granada (UGR)** | *Spain*
M.S., Applied Statistics

Jul. 2020 | **Universidad Carlos III de Madrid (UC3M)** | *Spain*
B.S., Statistics and Business

RESEARCH INTERESTS

- Constraint learning and surrogate models: embedding machine learning models into optimization problems
- Uncertainty quantification: prediction intervals, quantile regression and conformal prediction
- Trustworthy AI/ML for power and energy systems

PUBLICATIONS

Journal Articles (9)

[1] **Integrated design and scheduling of hydrogen processes under uncertainty: a quantile neural network approach.**
Ghilardi, L. M., Patrón, G. D., **Alcántara, A.**, & Tsay, C. (2025). *Industrial & Engineering Chemistry Research.* doi:10.1021/acs.iecr.5c03288

[2] **Leveraging neural networks to optimize heliostat field aiming strategies in concentrating solar power tower plants.**
Alcántara, A., Diaz-Cachinero, P., Sánchez-González, A., & Ruiz, C. (2025). *Energy and AI.* doi:10.1016/j.egyai.2025.100520

[3] **A quantile neural network framework for two-stage stochastic optimization.**
Alcántara, A., Ruiz, C., & Tsay, C. (2025). *Expert Systems with Applications.* doi:10.1016/j.eswa.2025.127876

[4] **Optimal day-ahead offering strategy for large producers based on market price response learning.**
Alcántara, A., & Ruiz, C. (2024). *European Journal of Operational Research.* doi:10.1016/j.ejor.2024.06.038

[5] **A neural network-based distributional constraint learning methodology for mixed-integer stochastic optimization.**
Alcántara, A., & Ruiz, C. (2023). *Expert Systems with Applications.* doi:10.1016/j.eswa.2023.120895

[6] **On data-driven chance constraint learning for mixed-integer optimization problems.**
Alcántara, A., & Ruiz, C. (2023). *Applied Mathematical Modelling.* doi:10.1016/j.apm.2023.04.032

[7] **Deep neural networks for the quantile estimation of regional renewable energy production.**
Alcántara, A., Galván, I. M., & Aler, R. (2023). *Applied Intelligence.* doi:10.1007/s10489-022-03958-7

[8] **Pareto optimal prediction intervals with hypernetworks.**
Alcántara, A., Galván, I. M., & Aler, R. (2023). *Applied Soft Computing.* doi:10.1016/j.asoc.2022.109930

[9] **Direct estimation of prediction intervals for solar and wind regional energy forecasting with deep neural networks.**
Alcántara, A., Galván, I. M., & Aler, R. (2022). *Engineering Applications of Artificial Intelligence.* doi:10.1016/j.engappai.2022.105128

Working Papers (2)

[10] **Trustworthiness layer for foundation models in power systems: application to N-k contingency screening.**
Alcántara, A., & Chatzivasileiadis, S. (2026). *arXiv:2602.07995*

[11] **Optimal placement of wind farms via quantile constraint learning.**
Feng, W., **Alcántara, A.**, & Ruiz, C. (2025). *arXiv:2510.01093*

TEACHING EXPERIENCE

Fall 2026 | **Data Analysis** — *CUNEF*
B.S. in International Business and Economics (1 group)
Main lecturer

2022–2025 | **Statistics for Engineering** — *UC3M*
B.S. in Mechanical, Electrical, Industrial and Aerospace Engineering (4 groups)
Co-course responsible for lectures, exam design and grading
Student evaluations: 4.38, 4.52, 4.75 and 4.92 (out of 5)

2023–2024 | **Introduction to Statistical Modeling** — *UC3M*
B.S. in Data Science and Communication Engineering (2 groups)
Co-course responsible for lectures, exam design and grading
Student evaluations: 4.92 and 5.00 (out of 5)

GRANTS & AWARDS

2021–2025 | FPU Fellowship, Spanish Ministry of Science and Universities
Highly competitive national fellowship (15 awarded nationally in field)

2022–2024 | Teaching Excellence Acknowledgement, UC3M

2023 | Competitive Mobility Grant, UC3M

2020 | Extraordinary End of Studies Award, UC3M

2019 | Social Council Excellence Award, UC3M

2019	Santander Progreso Scholarship, Santander Bank
2018	Excellence Award, Regional Government of Madrid

SELECTED TALKS

Invited Seminars

Apr. 2026 — GridFM Working Group
Trustworthiness Layer for Foundation Models in Power Systems.

Mar. 2026 — International System Operator Network (ISON)
Trustworthiness Layer for Foundation Models in Power Systems.

Feb. 2026 — CUNEF Universidad, Quantitative Methods Department, Spain
Embedding Machine Learning into Optimization: Strategic Decision-making in Energy Markets.

Oct. 2024 — Linköping University, Dept. of Mathematics, Sweden
A Quantile Neural Network Framework for Two-stage Stochastic Optimization.

Sep. 2024 — KTH Royal Institute of Technology, Dept. of Mathematics, Sweden
A Quantile Neural Network Framework for Two-stage Stochastic Optimization.

Sep. 2023 — Imperial College London, Dept. of Computing, United Kingdom
Probabilistic Forecasting, Quantile Neural Networks, and Stochastic Optimization.

Conference Presentations

Jul. 2026 — 24th Conference of the International Federation of Operational Research Societies (IFORS), Vienna, Austria
Leveraging neural networks to optimize heliostat field aiming strategies in Concentrating Solar Power Tower plants.

Jul. 2025 — 41st National Congress of Statistics and OR (SEIO), Lleida, Spain
A Quantile Neural Network Framework for Two-stage Stochastic Optimization.

Jul. 2025 — 34th European Conference on Operational Research (EURO), Leeds, United Kingdom
From Sample Average Approximation to Surrogate Models in Smart Grids Operation.

Jul. 2024 — 33rd European Conference on Operational Research (EURO), Copenhagen, Denmark
A Quantile Neural Network Framework for Two-stage Stochastic Optimization: Application to Power Systems Operation.

Jun. 2023 — 20th Int. Conf. on Constraint Programming, AI and OR (CPAIOR), Nice, France
A Neural Network-Based Distributional Constraint Learning Methodology for Mixed-Integer Stochastic Optimization.

FUNDED PROJECTS

2025 – Present — *ODEON – Federated Data and Intelligence Orchestration & Sharing for the Digital Energy TransitioN*
Horizon Europe · Role: Researcher

2024 – Present — *Algorithms for Stochastic Optimization Using Data-driven and Learning Analysis*
Spanish State Research Agency (PID2023-151013NB-I00) · Role: Researcher

2024 – Present — *SOLAROPIA-CM-UC3M – Operación e Integración de las Plantas Solares de Torre en el Sistema Eléctrico Español*
Comunidad de Madrid · Role: Researcher

2021–2024 — *Optimization under Uncertainty and Stochastic Control*
Spanish State Research Agency (PID2020-116694GB-I00) · Role: Researcher

2021 — *Probabilistic Forecasting and Meta-heuristic Optimization for Solar/Wind Resources in the Iberian Peninsula*
Spanish State Research Agency (PID2019-107455RB-C22) · Role: Research Assistant

SOFTWARE

CONDUCTOR — github.com/antonioalcantaramata/CONDUCTOR · Main developer, 2026
LLM-orchestrated digital twin for uncertainty-aware power-system operations, pairing a pandapower/Pyomo analysis backend with a natural-language agent.

Quantile2SP — github.com/antonioalcantaramata/Quantile2SP · 2025
Quantile neural network surrogates for two-stage stochastic programs.

CCL-Tool	github.com/antonioalcantaramata/ccl_tool · 2023 Chance-constraint learning and its integration with mixed-integer optimization.
POPI-HN	github.com/antonioalcantaramata/POPI-HN · 2022 PyTorch framework for hypernetwork-based Pareto optimal prediction intervals.

LANGUAGES & COMPUTER SKILLS

Languages	Spanish (native), English (fluent), Italian (basic)
Programming	Python, R, MATLAB
Libraries	PyTorch, scikit-learn, Gurobi, Pyomo, pandapower